Geophysics III

Algebra Tutorial

1 Introduction

Algebra is a set of rules and theorems that govern a broad swath of mathematics. While we'll employ calculus as well, much of what we do will be substitution and rearangement of abstract algebraic expressions. Abstract algebra simply means the use of symbols to represent numbers and concepts rather than numbers themselves. We'll use this absract representation so that we can learn about the general patterns rather than the solution for any specific case. And therein lies the power of abstract algebra: the ability to write a general expression applicable to many, perhaps infinitely many, cases.

This tutorial is meant to be a reminder of basic algebraic functions, properties, and techniques. If you have never had math beyond algebra, it may introduce a few new ideas and concepts that you weren't exposed, but are well within the confines of algebra. To some this may seem trivial, but to others it will be a useful review.

2 Rearranging equations

2.1 Rearranging rules

There are a few simple rules that you must remember when rearranging equations. You'll use them frequently when performing derivations. The rules:

1. perform the same operation to both sides of the equation;
2. when adding or subtracting, make sure the sum is zero since adding zero doesn't change a result; and
3. when multiplying or dividing, make sure the product is one because multiplying anything leaves the result unchanged;
4. when raising an exponent or taking a root, make sure the power is one.

Practice:
Show $2r(x^2-1) = x^2-2$ can be written as

$$x^2 = \frac{2(1-r)}{(1-2r)}. \tag{1}$$

2.2 Common mistakes

Most of the common errors occur when dealing with fractions. Although the rules of dealing with fractions are quite simple, the errors seem to happen because of temptation; temptation to cancel terms that simplify the expression, but generally too much. If you are ever unsure about a rule, TEST IT! Put real numbers in and test the before and after (use simple numbers). If the before and after don't give the same result, then what you did is invalid. If it does, try another set and if it still works then you probably did it right. I'll give a few examples of things that I commonly see but are incorrect.

Example 2.1. Mistakes commonly occur when trying to cancel terms that show up in both the numerator and denominator. For instance,

$$\frac{\cancel{z^2}}{x^2+\cancel{z^2}} \neq \frac{1}{x^2},$$

where the z^2 terms cannot be canceled. If we put in some numbers, 1 for x and 2 for z,

$$\begin{aligned}\frac{2^2}{1^2+2^2} &\neq \frac{1}{1^2}\\ \frac{4}{5} &\neq 1.\end{aligned}$$

It turns out that the above expression cannot be simplified further

We cannot this way either

$$\frac{x^2+\cancel{z^2}}{\cancel{z^2}} \neq x^2,$$

which we can test as above,

$$\begin{aligned}\frac{1^2+2^2}{2^2} &\neq 1^2\\ \frac{5}{4} &\neq 1.\end{aligned}$$

You can partially cancel the above expression,

$$\begin{aligned}\frac{x^2+z^2}{z^2} &= \frac{x^2}{z^2}+\cancelto{1}{\frac{z^2}{z^2}}\\ &= \frac{x^2}{z^2}+1\end{aligned}$$

Using our inputs above you can verify that the result is $1+\frac{1}{4}=\frac{5}{4}$.

Example 2.2.

$$\frac{x^2-2z^2}{x^2}-\frac{x^2}{z^2} \neq \frac{-2z^2}{x^2}$$

2.3 Rules for special functions

There are a number of handy properties of logs and exponents that seem to be commonly forgotten, which I don't blame you for. It is quite likely that you haven't used them since you learned them the first time. Turns out, we use them in science quite frequently; although, if you have only ever been given equations you my not realize that it has been done. These rules can be useful for simplifying equations or reorganizing them when performing derivations.

2.3.1 Logarithms

We can write a logarithm generally as

$$\log_{\underbrace{a}_{\text{base}}} \overbrace{u}^{\text{argument}}.$$

Although the logarithm above is written generically, the bases used is in most often in physics are either 10 or e, the natural number. For $\log_1 0$, we often use a shorthand Log and sometimes log. For $\log_e$, the shorthand ln or log are often used. Because log is used as the shorthand for both bases 10 and e in different texts, you will need to identify which base log applies.

Properties of logarithms, $u, v > 0$,

$$\begin{aligned} \log_a a &= 1 \\ \log_a a^n &= n \\ \log_a u^n &= n \log_a u \\ \log_a uv &= \log_a u + \log_a v \\ \log_a \frac{u}{v} &= \log_a u - \log_a v \end{aligned}$$

To change base, i.e., from a to b (where $a, b \neq 1$),

$$\log_b u = \frac{\log_a u}{\log_a b}.$$

There are several key points on a logarithm that are useful to know,

$$\begin{aligned} \mathrm{Log}\, 0 &= -\infty & \qquad \log 0 &= -\infty \\ \mathrm{Log}\, 1 &= 0 & \log 1 &= 0 \\ \mathrm{Log}\, 10 &= 1 & \log e &= 1. \end{aligned}$$

2.4 Exponentials

Hidden in the relationships above is the relationship between logarithms and exponentials as the inverse operations of each other, i.e.,

$$\begin{aligned} \log_a a^n &= n \\ a^{\log_a n} &= n. \end{aligned}$$

Properties of exponentials,

$$\begin{aligned} a^m a^n &= a^{m+n} \\ \frac{a^m}{a^n} &= a^{m-n} \\ (a^m)^n &= a^{mn} \\ a^{-n} &= \frac{1}{a^n} \\ a^0 &= 0 \\ a^1 &= a. \end{aligned}$$

If a and n are positive, $a^{\frac{1}{n}} = \sqrt[n]{a}$.

One must be carful with negative powers when a is also negative as they may yield a complex rather than real number (see complex number tutorial).

2.5 Simplifying solutions

We will try to write our solutions in as simple and elegant a form as possible. One reason to do so is that it can be easier to interpret, compute, and/or graph. But what one decides as an elegant representation is subject to debate. Also, it may be easier for a computer to compute a less elegant representation. Nevertheless, we will strive to do so. Here are a few suggestions to derive elegant solutions:

- combine like terms, no use carrying around the extra weight, e.g., $3x + x = 4x$;
- reduce fractions, again reduces extra weight, e.g., $\frac{6x^2}{64x} = \frac{3x}{32}$;
- find least common multiples, this can help us simplify a solution, e.g., $\frac{1}{x_1} + \frac{1}{x_2} = \frac{x_2+x_1}{x_1 x_2}$;
- factorize polynomials,, doing so can help us quickly identify roots (zero crossings), e.g., $\rho_1 r - \rho_2 r = (\rho_1 - \rho_2) r$; and
- expand/distribute (opposite of factoring), sometimes this simplifies the expression, e.g., $3z^2 - (x^2 - z^2) = 4z^2 - x^2$; and
- write linear relationships in slope-intercept form—the usefulness speaks for itself—or non-linear equations in a standard form, e.g., Pythagorean equation.

Also, for equations containing complex numbers, it is best to write in a form without complex numbers in the denominator, e.g., $\frac{1}{1-i} = \frac{1}{1-i}\frac{1+i}{1+i} = \frac{1+i}{2}$.

Since some of the above suggestions for simplifying equations are contradictory, *how will you know what I'm looking for as a grader?* The key is having all the correct elements so there is a fair amount of latitude to your answer so long as it has been simplified.

2.6 Practice problems

Problem 2.1. If

$$\sin\theta = \frac{x}{h},$$

show

$$\cos\theta = 1 - \frac{x^2}{h^2}.$$

You'll need to use the Pythagorean Theorem with respect to the sides of a right triangle with one of the two remaining angles θ.

Problem 2.2. Show that

$$y = 2\left(\frac{1}{x_1^2} - \frac{1}{x_2^2}\right)^{1/2}$$

is equivalent to

$$y = \frac{2(x_2^2 - x_1^2)^{1/2}}{x_1 x_2}.$$

Problem 2.3. Show that

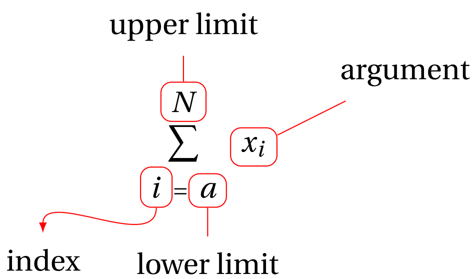
$$b = a_1^{v_1} a_2^{v_2}$$

is equivalent to

$$\log b = v_1 \log a_1 + v_2 \log a_2.$$

3 Special operators

3.1 Summation, $\sum$



At times we will want to add or multiply a known, and regular set of discrete numbers (note this will eventually lead to calculus). These will show up from time to time throughout the course. For example, suppose we wanted to add the integers from 1 to 5 whose result is 15,

$$1 + 2 + 3 + 4 + 5 = 15. \tag{2}$$

Because mathematicians are lazy, they seek a shorthand to write this in a simpler way. Rather than add up all the numbers at once, let's group these numbers to perform only one operation at a time, i.e.,

$$((((1) + 2) + 3) + 4) + 5. \tag{3}$$

Working with the inner most parentheses first, we find 1; inside the next set we find $(1+2)$, which yields 3. Continuing each step, we finally arrive at 15. This is the point where the simplifying notation comes in. Because each of the above operations is an addition, we use the capital Greek letter sigma (Σ) to denote the sum. We also give the sum limits: where to start (above) and where to end (below). Thus, we can write the sum as

$$\sum_{i=1}^{5} i \quad \text{where } i = 1, 2, \ldots, 5, \tag{4}$$

In this case, we perform the summation over the values of the index, i, from 1 to 5, incrementing i by a single value each step.

The increments of the index, need not be unity, but could be anything we need them to be so long it is regular or defined in some ordered way. The argument, also i in the example above, could also be anything.

We can express the arithemetic mean using the summation. For a set of N numbers, the arithmetic mean is given as

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_N}{N}, \tag{5}$$

or in short form,

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i. \tag{6}$$

Likewise, the sample standard deviation is defined as

$$\sigma_x = \left[\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2\right]^{1/2}. \tag{7}$$

Practice:
Prove to yourself

$$\sum_{i=1}^{4} i^2 = 30 \tag{8}$$

3.2 Product, $\prod$

There is an analagous operator for multiplication, just as we used the Σ for summation, the capital Greek letter pi (Π) is used for the product of a sequence. For example, we can write the product of N numbers as

$$\prod_{i=1}^{N} x_i = x_1 \cdot x_2 \cdot x_3, \ldots, x_N. \tag{9}$$

This notation arises when we use a geometric mean

$$\bar{x}_g = \left(\prod_{i=1}^{N} x_i\right)^{1/N}. \tag{10}$$

Practice:
Prove to yourself

$$\prod_{i=1}^{4} i^2 = 576. \tag{11}$$

4 Common functions

There are a number of functions that are quite common (Figure 1). We will use nearly all of these functions at least once if not repeatedly within the course. It is extremely important to know what these functions generally look like and know their attributes. The attributes that are the most important to understand regard how they can be manipulated to fit physical models and/or data and provide insight into the physical processes that we use them to describe. These attributes include their scale, amplitude, slope, and key points: location of asymptotes, zero crossings, intercepts etc.

4.1 Linear

A line is one of the simplest functions (and most useful) functions that we can use in any science. A linear process is one in which one variable is proportional to another by a constant. This constant provides the scale—commonly know as the slope. Add an additional constant that accounts for a

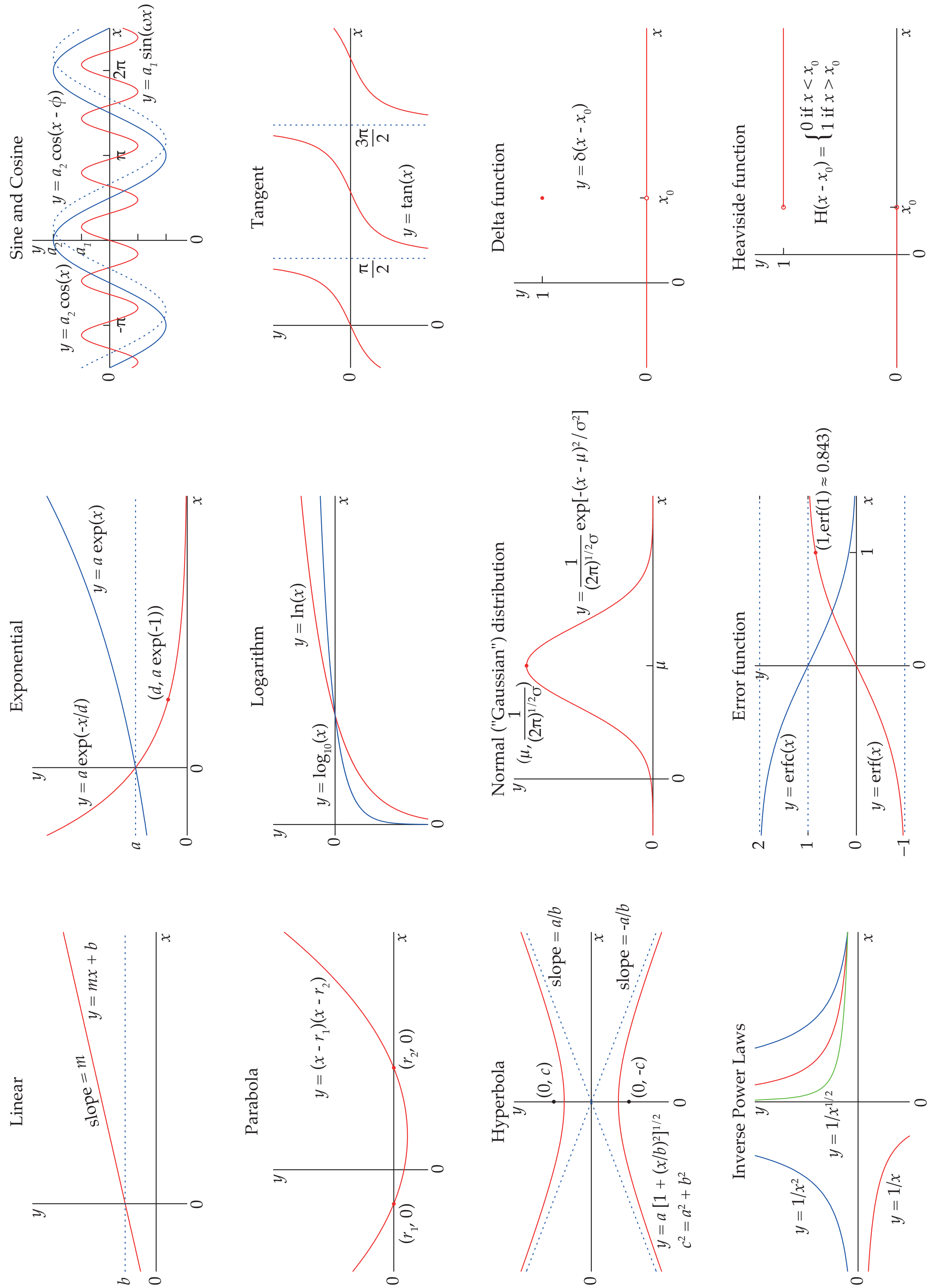


Figure 1: Common functions.

shift or static offset between each of the variables (the intercept). The equation of a line is given by

$$y = mx + b, \tag{12}$$

where m is the slope and b is the intercept. Depending on our process the slope and/or intercept may be represented by multiple physical constants that combine to make a single constant. An alternative form of the linear equation that is at times handy can be written,

$$y - y_0 = m(x - x_0). \tag{13}$$

A line also represents a first order, and the simplest, polynomial. You will see lines in many places throughout geophysics.

4.2 Parabola/Quadratic

A parabola is a second order polynomial, written

$$y = a_2x^2 + a_1x + a_0, \tag{14}$$

with three constants, a_0, a_1, and a_2. You'll notice that the form I have written in Figure 1 as

$$y = (x - r_1)(x - r_2), \tag{15}$$

where r_1 and r_2 are the roots (zero crossings) of the parabola. While the equation can always be written in this form, r_1 and r_2 are only real valued when the equations intersect the x-axis. When the parabola does not intersect the x-axis, the roots are complex numbers (we'll get to complex numbers later). To find the roots of a parabola, we can use the famous quadratic formula,

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{16}$$

You'll see a quadratic when we get to heat flow.

4.3 Hyperbola

A hyperbola is a conic section (as are parabolas, circles, and ellipses). They are also the shape of a seismic reflection. A hyperbola can be expressed in the form

$$y = \pm a\left[1 + \frac{x^2}{b^2}\right]^{1/2}, \tag{17}$$

where a and b are constants. A hyperbola has two asymptotes, each with a slope defined by the ratio of the constants, i.e., $m = \pm\frac{a}{b}$. The foci are also defined by the constants, $c^2 = a^2 + b^2$. The y-axis intercept of the ellipse is defined by the constant, a.

4.4 Inverse Power Laws

Functions which are represented by the inverse of a power,

$$y = x^{-n}, \tag{18}$$

where n is a positive multiple of $1/2$ are relatively common. For instance, gravitational acceleration is proportional to the inverse square of the distance between two bodies, i.e., $\propto r^{-2}$. As $x \to 0$, the function grows to infinite magnitude, and as $x \to \infty$, the function approaches zero.

Now it is your turn. For the remainder of these, I'll let you have some practice writing up your own observations. I suggest you graph these functions. Change some of the constants and see how it changes the graph. What attribute(s) did it affect?

Remember, these functions are used to describe physical processes and many of the key properties can be determined directly from the function and/or graph.

4.5 Exponentals

Where might these be common? What is the general form? Are there any key points, asymptotes, discontinuities?

4.6 Logarithm

4.7 Gaussian

4.8 Error function

4.9 Trigonometric functions

4.9.1 Sine

4.9.2 Cosine

4.9.3 Tangent

4.10 Delta function

A delta function is expresed as

$$y = \delta(x - x_0). \tag{19}$$

The delta function is a special function that is always zero except when $x = x_0$ at which point the function is unity. While a bit strange, these are very useful functions when we work with waves, but they are used in many engineering and physical fields.

4.11 Heaviside function